Int. J. Production Economics 224 (2020) 107548

Contents lists available at ScienceDirect

International Journal of Production Economics

journal homepage: http://www.elsevier.com/locate/ijpe

Bullwhip effect in the oil and gas supply chain: A multiple-case study

Tianyuan Zhu [a,*], Jaydeep Balakrishnan [a,b], Giovani J.C. da Silveira [a]

[a] Haskayne School of Business, University of Calgary, 2500 University Dr NW, Calgary, Alberta, T2N 1N4, Canada
[b] College of Business Administration, California State University, 6000 J Street, Sacramento, CA 95819, USA

ARTICLE INFO

Keywords:
Bullwhip effect
Oil and gas
Energy
Case study
Supply chain management

ABSTRACT

The bullwhip effect has been extensively studied in the retail, wholesale and manufacturing industries. However, it has been rarely explored in the context of resource extraction industries such as oil and gas, despite their economic impact and distinct features. This paper investigates the factors that impact the bullwhip effect in the oil and gas supply chain using case study evidence from six companies in North America, which cover refining and marketing, exploration and production, integrated oil and gas, and drilling. For each type of company studied, the operational causes of the bullwhip effect proposed in the literature and other factors of influence are examined. The findings indicate that the existing theories of the bullwhip effect have limitations in explaining the phenomenon in the oil and gas industry. Information sharing, a widely advocated countermeasure of the bullwhip effect may not be relevant in the integrated oil and gas company. Regarding the factors that drive or mitigate the bullwhip effect in different types of companies in the oil and gas supply chain, seven propositions are developed and several additional findings are obtained. All of these results enable better understandings of the bullwhip effect in academia, oil and gas organisations and related industries, and may provide guidance for potential countermeasures in practice.

1. Introduction

As one of the most celebrated concepts in the operations management field, the bullwhip effect (BWE, also known as "demand amplification" or "variance amplification") denotes the phenomenon where demand variability is amplified upstream in the supply chain. This phenomenon was first identified in the context of system dynamics by Forrester (1958, 1961). Subsequently, Sterman (1989) extended Forrester's model and studied the phenomenon through the well-known "Beer Distribution Game". The term "bullwhip effect" was coined and introduced to the literature by Lee et al. (1997a, 1997b), which then stimulated multiple streams of research. In the past 20 years, nearly 600 research papers related to the BWE have been published (Wang and Disney, 2016). Among these studies, some researchers measure the BWE theoretically (e.g., Chen et al., 2000; Gao et al., 2017; Kadivar and Akbarpour Shirazi, 2018; Wang et al., 2016; Zhang, 2004) or in empirical contexts (e.g., Bray and Mendelson, 2012; Isaksson and Seifert, 2016; Pastore et al., 2017; Shan et al., 2014). Having observed the discrepancy between theoretical and empirical BWE values, Trapero and Pedregal (2016) proposed a novel dynamic BWE metric to bridge the gap between these two types of measures. As the BWE can create significant inefficiencies in the supply chain (e.g., excessive inventories, misguided production or capacity planning, poor customer service and lost revenue due to shortages), many researchers have identified the causes for the BWE and proposed a variety of remedies (e.g., Croson and Donohue, 2006; Dejonckheere et al., 2003; Disney and Towill, 2003; Haines et al., 2017; McCullen and Towill, 2002; Zhang and Burke, 2011). Comprehensive reviews of the BWE literature can be found in Geary et al. (2006), Miragliotta (2006) and Wang and Disney (2016).

Existing BWE studies mainly focus on the retail, wholesale and manufacturing industries, while very few have investigated the resource extraction industries, such as the oil and gas (O&G), which are at the upstream end of the supply chain. To the best of our knowledge, the earliest evidence of BWE in the O&G supply chain was provided by Bishop et al. (1984). They described how the fluctuations in demand for O&G caused dramatic changes in demand for turbomachinery directly involved in O&G extraction during the 1970s. Sterman (2006) mentioned that O&G drilling activity fluctuated about three times more than the production between 1975 and 2005, which generated large oscillations in supplies of drilling rigs and oilfield services. Jacoby (2010) also proposed that the swings in O&G producers' drilling investment led to even bigger swings in the equipment supply chain. In the

* Corresponding author.
E-mail addresses: zhut@ucalgary.ca (T. Zhu), jaydeep.balakrishnan@csus.edu (J. Balakrishnan), giovani.dasilveira@haskayne.ucalgary.ca (G.J.C. da Silveira).

https://doi.org/10.1016/j.ijpe.2019.107548
Received 5 October 2018; Received in revised form 6 August 2019; Accepted 30 October 2019
Available online 4 November 2019
0925-5273/© 2019 Elsevier B.V. All rights reserved.

prevailing works on the BWE, some empirical studies have shown the existence of BWE in the O&G supply chain. Using the U.S. industry-level data, Cachon et al. (2007) demonstrated that the variance of material inflow is greater than the variance of sales in industries of petroleum and petroleum products wholesale, petroleum and coal products manufacturing, as well as mining and O&G field machinery manufacturing. The firm-level mean overall BWE reported by Bray and Mendelson (2012) showed that orders in the O&G extraction industry are 14.64% more variable than demands, while in petroleum and coal manufacturing, the orders are only 4.28% more variable than demands. More recently, Shan et al. (2014) reported the existence of BWE in the energy sector in China. Isaksson and Seifert (2016) quantified the BWE using two-echelon data (buyer-supplier dyads). They observed a significant BWE in the O&G extraction industry supported by several robustness checks. Additionally, results obtained by Bray and Mendelson (2012) and Isaksson and Seifert (2016) indicated that the O&G extraction industry has the highest BWE in the resource extraction sector.

Although progress has been seen in renewables, hydroelectric and nuclear power in the past decades, nearly 60% of the world's primary energy today still comes from oil and natural gas (BP, 2018). The O&G industry also touches everyone' lives with various products such as asphalt, lubricants and petrochemical products (e.g., eyeglasses, clothing), and impacts national security, the global economy and politics (Inkpen and Moffett, 2011). Besides its significance, the O&G supply chain also has characteristics that are different from the supply chains explored in most BWE studies. First, the major players in the O&G supply chain, such as crude oil and natural gas producers, refining companies and drilling companies are all capital intensive. Second, other than products, companies in the O&G supply chain also buy or sell services. For example, O&G producers sell crude oil to refining companies, while drilling companies and oilfield service companies provide well drilling and setting up services to O&G producers. Further, since O&G are commodities, O&G producers are price takers. Also, crude oil and natural gas prices are inherently volatile and difficult to predict. Moreover, both O&G producers and refining companies have high-volume operations and continuous processes. Finally, pipeline systems are used in crude, natural gas and refined petroleum transportation.

Due to these differences, it is not clear whether the existing theories of BWE can apply to the O&G industry, and in particular, what factors may increase or attenuate the BWE in the O&G supply chain. In this research, we attempt to address these questions through a multiple-case study in the O&G industry. More specifically, we would like to investigate (1) whether the operational causes of the BWE proposed in the prevailing studies exist in different types of companies in the O&G supply chain; (2) whether other factors may cause or smooth the BWE in different types of companies in the O&G supply chain.

The remainder of the paper is structured as follows: The operational causes of the BWE proposed in the existing literature are briefly reviewed in Section 2. In Section 3, we first identify the O&G supply chain and describe the different types of companies; then, discuss the features of the O&G industry and their implications for the understanding of the existence of BWE in the O&G supply chain. Section 4 introduces the research methodology. This is followed by the case analysis in Section 5. In Section 6, we discuss our findings and the generalisation of the results. Finally, Section 7 concludes our research, discusses limitations and suggests avenues for future research.

2. Operational causes of the BWE

The causes for the BWE can be divided into two categories: operational and behavioural. Behavioural causes follow the bounded rationality of decision-makers, whereas operational causes explain the BWE as the consequence of rational reactions to those well-perceived factors. Lee et al. (1997a, 1997b) pointed out five major operational causes of the BWE: demand forecast updating, non-zero lead time, order batching, price fluctuations and rationing and shortage gaming.

Demand forecast updating increases the BWE when demand is nonstationary, and the company uses historical demand from immediate rather than end customers to update forecasts. Thus, a surge of demand in one period will be interpreted as high demand in the future, and subsequently result in higher order quantities. Variations in order quantities are further amplified with longer lead times and more intermediaries in the supply chain, since target inventory level increases with the lead time, and every intermediary adds safety buffers. Apart from duration, lead time variability can also increase the BWE (e.g., Ancarani et al., 2013; Chatfield et al., 2004; Duc et al., 2008; Kim et al., 2006). Order batching occurs mainly due to economies of scale (e.g., reducing order transaction costs, obtaining volume discounts and achieving full truck shipments), but may also result from a periodic inventory review process. Price fluctuations are usually generated by special discounts or promotions. Attractive pricing offered by the manufacturer or distributor often leads to forward buying in the downstream. As downstream actors concentrate orders and build up stocks when the price is low, the demand pattern is distorted, increasing the BWE. When demand exceeds supply, the manufacturer usually allocates supply in proportion to the amount ordered. Recognising this rationing strategy, customers tend to artificially inflate orders to secure resources and satisfy true demand, and later cancel any excessive orders. As a result, the manufacturer has an inaccurate perception of actual demand, which may generate excessive inventory and production capacity when the supply becomes sufficient, and customers no longer inflate their orders.

In addition to the causes proposed by Lee et al. (1997a, 1997b), multiplier effect, also known as the "investment accelerator effect" in macroeconomics, is a cause of the BWE often overlooked by researchers. It refers to the phenomenon that a small change in demand for consumer goods creates a dramatic change in demand for capital equipment used to produce those goods (Anderson et al., 2000; Wang and Disney, 2016). Different from finished goods whose residence time in inventory is rather short, capital equipment is typically replaced on a depreciation basis, whose residence time is its lifetime (Anderson and Fine, 1999; Anderson et al., 2000). For example, suppose that a firm is operating at full capacity with 20 machines, and replaces its machine at the rate of 10% per year (i.e., the average machine lifetime is ten years). In this case, the quantity of machine ordered is generally stable (i.e., two machines per year). However, if the company expects a sustained 5% increase in demand, they will purchase machines to expand its production capacity by 5%. Thus, the machine order quantity increases to three in that year (a 50% increase), and the change in product demand (5%) is amplified ten times, causing the BWE. However, it should be noted that this example is oversimplified. In practice, the constant capital output ratio (e.g., one machine produces one product per quarter) may not necessarily hold constant; varying depreciation policies can affect the replacement demand; for new equipment, replacement demand may not exist in the short run; and obsolescence also influences the order quantity (Bishop et al., 1984). Bishop et al. (1984) described how the multiplier effect (they called "the acceleration principle") and purchasing agents' behaviour cause volatility in demand for turbomachinery equipment used in O&G production. Anderson et al. (2000) demonstrated that the multiplier effect is an important source of the BWE in the machine tool supply chain.

Aside from the causes above, Lee et al. (1997b) stated that the production capacity limitation might cause BWE through rationing and shortage gaming. De Souza et al. (2000) and Paik and Bagchi (2007) also demonstrated that production capacity limitations could contribute to the BWE using the beer game model. Taylor (1999, 2000) suggested that supply variability is another possible cause of the BWE, which includes "variability in machine reliability and output" and "variability in process capability and subsequent product quality". These problems can cause intracompany problems and uncertainties among companies,

which in turn, may affect each company's behaviour and lead to order variability (Moyaux et al., 2007). For a more thorough review of the causes for the BWE, we refer the reader to Bhattacharya and Bandyopadhyay (2011).

3. The O&G industry

3.1. The O&G supply chain

To study the BWE in any context, the first step is to identify the supply chain echelons. The O&G supply chain starts from exploration and production of crude oil and natural gas, goes through refining, and ends with distribution and delivery of the final products to customers. It has four main echelons, which are illustrated in Fig. 1. From downstream to upstream, the first tier is the O&G refining and marketing company, which processes crude oil, sells refined products (e.g., gasoline, diesel, distillates), and/or commercialises and distributes natural gas. O&G exploration and production (E&P) companies, in Tier 2, explore and produce crude oil and natural gas. Companies who engage in both Tier 1 and Tier 2 (or even more tiers) are called integrated O&G companies. These companies play a significant role in the petroleum industry, especially some large international oil companies (IOCs) such as Shell, and national oil companies (NOCs) like PetroChina. O&G storage and transportation companies store and transport crude oil, natural gas and refined products; some also engage in trading (e.g., Enbridge). Even though O&G storage and transportation companies are often regarded as the midstream companies between E&P companies (upstream) and refining companies (downstream), we do not consider them as a separate echelon in this study, since we examine the phenomenon of demand amplification, whereas these companies generally do not buy and sell.

In O&G extraction, the E&P company first conducts a seismic survey on a property; then, drills exploratory wells to gather more detailed geological data and locate proven reserves of recoverable O&G. If oil or gas is discovered, development wells will eventually be drilled. After the well completion process is finished, the well can start producing oil or gas. Hence, the third tier in the supply chain generally includes drilling contractors as well as property evaluation and oilfield service providers, while companies in Tier 4 provide essential equipment, supplies and services to companies in Tier 3. For companies in Tier 3, we only focus on drilling contractors as drilling is the core activity. Companies in Tier 4 are mainly manufacturers which have been widely studied in existing literature, thus are not included in this research.

In summary, we will investigate four types of companies in three echelons within the O&G supply chain, including refining and marketing, E&P, integrated O&G and drilling.

3.2. Features of the O&G industry

Companies in the O&G supply chain studied in this research are all characterised by capital-intensive development. Large capital investments are required for refinery construction, and drilling contractors invest heavily in drilling rig fleets. In the E&P and upstream section of the integrated O&G companies, much capital is spent on exploration and well development, which includes the costs of well drilling services ordered from drilling contractors. Every year, E&P and integrated O&G companies set specific capital expenditure budgets for exploration and drilling activities. Drilling companies also set annual budgets for investments in drilling rig equipment. This is different from the traditional procurement in manufacturing, wholesale and retail supply chains, which depends on demand forecasting and order-up-to policies. Since drilling rigs rented out to E&P companies (or the upstream section of integrated companies) include specialised labour, drilling companies operate as service providers. Because services cannot be stored or rationed, it is expected that traditional BWE causes such as order batching, forward buying due to price fluctuations, and shortage gaming will not exist in contracting drilling rigs.

Another important aspect is that crude oil and natural gas are traded in commodity markets, which are liquid and include many sellers and buyers (Kurian, 2013). Depending on its viscosity and sulfur content, crude oils have different grades and prices. However, within each grade, the product cannot be further differentiated. Natural gas is also an undifferentiated product. As there are abundant volume and many suppliers, and customers are indifferent to the product supplier, there should be little to no rationing and shortage gaming between suppliers and buyers.

Since crude oil and natural gas are traded as commodities, O&G producers are price takers. Crude oil prices are driven by various factors including international and domestic supply and demand, macroeconomics, geopolitics, regulations, actions of the Organization of the Petroleum Exporting Countries (OPEC), weather, transportation and other infrastructure constraints. Changes in these factors may lead to rapid and significant fluctuations in oil prices (see Fig. 2). For example, in 1990, crude oil production in Iraq and Kuwait, accounting for nearly 9% of the world's total at that time, collapsed during the Persian Gulf War, and there were also threats to the production in Saudi Arabia (Hamilton, 2011). In consequence, crude oil prices doubled in three months, but dropped back to pre-war levels five months later, since Saudi Arabia used its excess capacity to restore world production (Hamilton, 2011). More recently, strong growth in demand, particularly in China, and stagnant supply between 2005 and 2007 dramatically increased oil prices. However, with the global financial crisis, prices fell from historic highs of over $140 in July 2008 to around $35 in February 2009. Likewise, weakening global demand, combined with growth in the U.S. shale oil production and OPEC's decision to maintain output led to a drop from over $100 in mid-2014 to an average $50 in 2015 (Behar and Ritz, 2016; Reinhard et al., 2017). Due to these impacts, crude oil prices can be highly volatile and difficult to predict. Even though futures prices may indicate expectations, actual price prediction remains complex (Inkpen and Moffett, 2011), and the dramatic short-period changes can be unpredictable.

Based on the BWE theory, crude oil price volatility would likely stimulate forward buying behaviour by refining and marketing companies. Crude oil prices are also closely related to the revenue and profitability of E&P and upstream sections of integrated O&G companies. Since drilling activity planning depends on the company's budget for capital investment, it is expected that oil price volatility, especially the unpredictable oil shocks will increase the BWE in the E&P and upstream section of the integrated O&G company.

Furthermore, O&G producers and refineries have continuous operations, similar to the process industries such as chemicals and pharmaceuticals, and are different from the manufacturers (e.g., automobile assembly) commonly discussed in the BWE literature (Jacoby, 2012). The pipeline system used for oil, gas and petroleum products shipping is also a continuous process. These raise the question of whether the nature of continuous operations influences the BWE in the O&G supply chain.

Finally, shipping plans for pipeline transportation are made on a monthly basis. Before a shipping period starts, shippers need to provide the pipeline company with accurate nominations[1] for all the volumes they plan to move in that month. According to a 2018 report by the National Energy Board (NEB) in Canada, pipeline apportionment occurs when the total volume of crude oil nominated for transportation exceeds the available pipeline capacity (NEB, 2018). This is similar to the rationing strategy used by manufacturers when demand is greater than supply. In an attempt to obtain sufficient capacity in pipeline apportionment, some shippers may nominate more than the actual volume they intend to ship on a pipeline (NEB, 2018). Thus, it is possible that this kind of gaming in the pipeline system may increase the crude oil

[1] The volume of petroleum, the receipt and delivery points and the type(s) of petroleum specified by the shipper in the monthly notice of shipment.

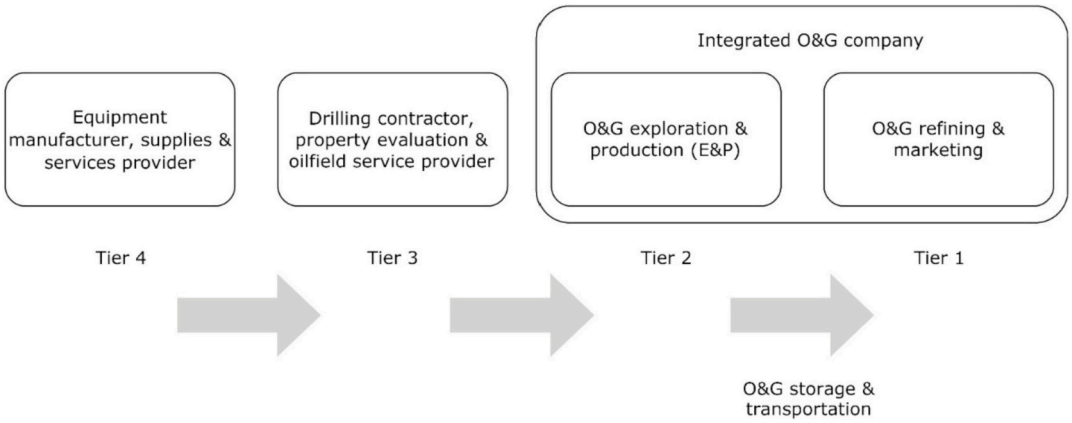

Fig. 1. O&G supply chain.

Source: Data retrieved from U.S. Energy Information Administration, Crude Oil Prices: Brent and WTI, https://www.eia.gov/dnav/pet/PET_PRI_SPT_S1_D.htm

Fig. 2. Crude oil prices from 1988 to 2018.

order variability, thereby intensifying the problem of BWE in the refining and marketing company.

The features discussed above make the O&G industry significantly different from industries traditionally studied in the BWE literature, calling for further investigation. Moreover, findings from the O&G industry in this study may also have implications for other resource extraction industries with similar characteristics.

4. Research methodology

As mentioned, the causes of the BWE proposed in the prevailing literature are all based on the knowledge and practice in retail, wholesale and manufacturing industries. Since the factors drive and smooth the BWE in the O&G the supply chain are unclear yet, we chose an inductive multiple-case study approach suggested by Eisenhardt (1989) and Yin (1994), which is appropriate for investigating underexplored contexts. The multiple-case study also allows to examine different types of companies in the O&G supply chain and reach more generic conclusions (Eisenhardt and Graebner, 2007).

Multiple-case study is inherently very time consuming due to its in-depth nature of the investigation. Thus, the sample size is limited, and the case selection is of fundamental importance in the research design (Dubois and Araujo, 2007). Our sample is composed of six companies in the O&G supply chain in North America. Two case companies were investigated in each of the three echelons defined in Section 3.1. According to McCutcheon and Meredith (1993, p. 243), for exploratory case studies focusing on theory development, such as this research, "It may be helpful to select several very different settings, through deliberate 'theoretical sampling' (Glaser and Strauss, 1967) that reflect the range of conditions thought to affect outcomes. Commonalities and differences across the varied settings help to outline the patterns upon which to develop theory." Eisenhardt (1989) stated that given the limited number of cases that can be investigated in one study, it makes sense to choose cases which are in extreme situations or provide examples of polar types. Following these guidelines, we selected our cases as follows. From Section 3.1, both refining and marketing and E&P companies can be divided into two categories: non-integrated company and the section of an integrated company. Thus, for refining and marketing companies, we chose one pure refining and marketing company, and one refining and marketing section (i.e., downstream section) in an

integrated O&G company. Similarly, the two cases we selected for E&P companies include one pure E&P company and one E&P section (i.e., upstream section) in another integrated O&G company. For drilling contractors, we chose one privately held, small drilling company and a publicly held, large drilling company. Another key consideration in selecting our cases was the company's willingness to participate in this case study.

Each of the six case companies studied was labelled using a reference code to retain anonymity. Company RM1 is a pure refining and marketing company, owning one oil refinery in North America. RM2 is the refining and marketing section of an integrated O&G company, which has multiple oil refineries in Canada and the U.S. Company EP1 is a pure E&P company and a subsidiary of an overseas O&G company. EP1 has both onshore and offshore operations and owns assets globally. However, we only consider its onshore operations in North America to make it comparable to EP2, which is the E&P section of an integrated O&G company who does onshore drilling and production in Canada. Company DC1 is a privately held drilling contractor with a small rig fleet of only a few land-based drilling rigs operating in Canada. DC2 is a public drilling company, which has a large rig fleet (over 200) serving both Canada and the U.S. Table 1 presents a summary of the case companies' backgrounds and their respondents.

Executives from the six case companies were interviewed. While their titles may vary, the respondent position in each of the three echelons can be described as follows. In the refining and marketing company, the manager in the supply and marketing department who is involved in crude oil purchasing and refined products sales was contacted. Specifically, the manager of RM2 also engages in sales of crude oil. For E&P companies, we talked to the drilling and completion sourcing manager who is in charge of drilling activity planning, drilling rig contracting and operations of the drilling program. In drilling companies, the drilling rig contracts manager was interviewed. Particularly, the contracts managers in DC1 and DC2 have worked in their companies for about eight and 11 years, respectively. Hence, both of them also have rich knowledge about the operations and procurement in the drilling company.

To obtain in-depth data and detailed descriptions, the key data collection method employed was the semi-structured interview, which allowed the respondents to answer questions and provide explanations. The interview protocol is available from the authors upon request. Except for the manager from RM1, all other respondents attended two semi-structured interviews; each took about 60–90 min. Due to scheduling problems, the manager of RM1 chose to participate in four telephone interviews; each took about 30 min. One of the researchers was the only interviewer, which enables the activity being performed consistently across cases. In addition to the qualitative data, quantitative data not leading to company identification were also collected to support more detailed explanations and examples. The interviews were recorded and transcribed. After each interview, a summary was e-mailed to the participant to validate the content. Respondents were also contacted through email or telephone when clarification or supplementary information was needed.

Besides interviews, we also collected additional data from company documents and secondary sources. These included redacted internal reports, summaries, and company records, quarter and annual financial reports, information on firm websites, news, industry-specific databases, as well as publications and reports from relevant government authorities (e.g., NEB, U.S Energy Information Administration) and independent agencies (e.g., Alberta Energy Regulator (AER), U.S. Securities and Exchange Commission). The rich data allowed for data triangulation, thereby increasing the data internal consistency (Voss et al., 2002). The reliability of the study was further increased by creating a database for the interviews and case study documents and establishing a chain of evidence as described by Yin (1994).

Cases were analysed following the process suggested by Miles and Huberman (1994), including both within-case analysis and cross-case analysis. Specifically, cross-case comparisons were conducted for each pair of case companies in the same echelon to detect commonalities and differences in the factors causing or smoothing the BWE across different settings, which is helpful to outline patterns and develop theory (Eisenhardt and Graebner, 2007; McCutcheon and Meredith, 1993; Yin, 1994). Since the BWE may be caused by interactions between buyers and sellers (e.g., rationing and shortage gaming) and the integrated O&G company covers two echelons in the supply chain, cross-case analysis was also conducted between companies in adjacent echelons, so as to improve understanding of the BWE factors. Finally, we developed propositions following the iterative analytic process suggested by Eisenhardt (1989).

5. Case analysis and findings

Drawing on existing theory and evidence from the cases, in this section, we examine whether the causes proposed in the prevailing studies exist in the O&G supply chain, and develop propositions about the factors that influence the BWE for different types of companies. Seven propositions and additional findings are described in details in the following subsections.

5.1. Refining and marketing company

Contrary to the findings of Lee et al. (1997a, 1997b), the problem of demand forecast updating is not an issue in either of the two companies interviewed, albeit for different reasons.

Refining and marketing companies order crude oil on a monthly basis. The quantity of crude oil is measured by barrels. RM2 determines its crude oil order quantity based primarily on the demand forecast of refined products following the order-up-to policy. Refined products sales in RM2 can be divided into two parts: refined products sold through long-term contracts and refined products sold on the spot market. Typically, long-term contracts for major refined products such as gasoline and diesel are often one to two years, with a fixed delivery volume for each month. Hence, every month when the refining and marketing

Table 1
Case study companies and respondents.

Company Label	Type	Ownership	Respondent Position	Description
RM1	Pure refining and marketing	Public	Director of supply & trading	One oil refinery in North America
RM2	Refining and marketing section of integrated company	Public	Vice president of commodity supply & marketing	Multiple oil refineries in North America
EP1	Pure E&P company	Subsidiary	Program manager of strategic procurement	Onshore E&P operations in North America
EP2	E&P section of integrated company	Public	Manager of drilling & completion sourcing, and supply chain management	Onshore E&P operations in Canada
DC1	Drilling contractor	Private	Strategic partnerships manager	Drilling rig fleet size: small, operates in Canada
DC2	Drilling contractor	Public	Contracts manager	Drilling rig fleet size: large, operates in North America

company forecasts the future demand, part of it is known with certainty based on fixed delivery quantities in their signed long-term contracts. As the manager from RM2 noted, "We contract a significant volume of our refined product sales to buyers who must take, and marginal volumes are often priced daily based on very competitive supply-demand dynamics at truck rack terminals. […] During the recent five years, of our total sales, about 70% is sold under long-term contracts." Due to the high proportion of long-term contracts used in refined products sales, future demand is largely known in advance for RM2, while only a small fraction of demand needs to be predicted using historical data. The effect of demand forecast updating is thereby alleviated. This "greatly improves the certainty of crude oil demand", as the manager said. Variation of monthly crude oil order quantity is thus reduced. Hence, the following result is obtained:

Finding 1. Higher proportion of long-term contracts with fixed delivery quantities used in refined products sales increases certainty about future demand, thus lowering crude oil order variability.

As refined products are shipped out from the refinery through pipeline, demands come in and deplete the inventory all day every day. The demand quantity of refined products is also measured by barrels. Fluctuations are often introduced into the refined product demand by seasonality (e.g., summer holiday driving), competitor pricing, special weather (e.g., major snowstorm), and unplanned outage in another refinery in the same region (e.g., the massive supply disruption in Texas caused by Hurricane Harvey eventually led to a surge in demand for RM2's refineries in the U.S.). Knowing these, in RM2, for the part of refined products not contracted to long-term commitments, the demand forecast is not only based on historical monthly demand data but also other information which explains the patterns and fluctuations in demand such as seasonality, weather, daily prices published by other parties, and activity of other refineries in the same region. This is different from the time series methods commonly discussed in the existing literature (e.g., Li et al., 2014; Zhang, 2004).

The forecasting mechanism applied may contribute to the BWE, since the parameter used in the mechanism determines the relative proportion of past demand fed forward into the future (McCullen and Towill, 2002). For RM2, by considering additional information in the demand forecast, the company has a better understanding of the temporary demand increase (or decrease). Hence, with proper adjustments based on the additional data, the relative proportion of past demand fed forward into future periods they use in the forecasting should be less likely to cause overreacting and excessive inventory. As the manager remarked, "A surge of demand may lead to us expecting more demand, but seasonality, weather, other refinery activity may explain this, so we take many data points into account to get a better forecasting result." In this case, the effect of demand forecast updating is mitigated, thereby reducing the variation of monthly crude oil order quantity. Thus, the second insight from the refining and marketing company is:

Finding 2. In demand forecasting for refined products, appropriately using additional information that explains the demand fluctuations reduces crude oil order variability.

RM1's demands come from both domestic and overseas. Domestically, though there are some refined products imported, since RM1 has the only refinery in the local market, the competition is not that intense. Besides, RM1 has agreements to export its products. So, "there is always demand there if we want to sell to overseas in the spot market" as the manager said. Therefore, the market demand is always sufficient for RM1. On the other hand, however, the biggest challenge for RM1 is to get enough supply of crude oil, because the refinery is physically distant from the oilfield, the oil shipping pipeline is always oversubscribed, and buying extra pipeline capacity (i.e., the volume of crude oil can be shipped per day) from other shippers or using land transportation is much more expensive. In this case, though the company does demand forecasting, it is not decisive in determining the crude oil order quantity. Instead, the pipeline capacity is the major constraint and crucial factor. RM1 always orders as much as they can get through the pipeline, since the volumes they want are more than they can get allocated almost all the time, except when there is maintenance in the refinery. Due to the limited pipeline capacity, even if the company has purchased excessive crude oil, the extra volume is not very large, which can usually be used to produce refined products and sold to export customers instead of storing for later sales. Further, the pipeline capacity allocated to the company is generally stable. As the manager mentioned, "In the past four years, the pipeline capacity allocated usually fluctuated within ± 5%. Sometimes, it could change by up to ± 10%, but this was uncommon." Hence, the problem of demand forecast updating is eliminated in RM1 because of the limited and stable transportation capacity. The variation of monthly crude oil order quantity is therefore lowered. In contrast, if RM1 does not have those export agreements, or if they cannot make a profit from export, the manager stated that they would determine the crude oil order quantity based on demand forecast like other refining and marketing companies as well. This leads to the following proposition:

Proposition 1. *Limited and stable pipeline transportation capacity reduces crude oil order variability*.

The crude oil supply lead time for RM1 is about 40 days, and the typical lead time in RM2 is about 55 days, both considered long. According to Lee et al. (1997a, 1997b), long lead time aggravates the BWE caused by demand forecast updating. Since the demand forecast updating problem is largely mitigated in the two companies by (1) a high proportion of long-term sales contracts, (2) properly using additional information in demand forecast and (3) limited and stable transportation capacity, the impact of long lead time on the BWE is alleviated. Besides, since pipeline is a relatively reliable transportation mode, lead time variation is small, which also cannot significantly increase the BWE in the refining and marketing company.

Though refining and marketing companies review their inventories, make forecasts and determine crude oil order quantities on a monthly basis, they do not purchase the crude oil they need in large batches. This is because the price of crude oil fluctuates every day, and the company prefers to average out their purchases to smooth the exposure to commodity price risk. In this case, the company usually purchases a certain volume every day or every few days in the month before the crude oil is injected into the pipeline for transportation. Therefore, order batching is not a major cause of the BWE in refining and marketing companies. Since both the crude oil production process and pipeline transportation process are continuous, the production and delivery are evenly spread in a period of time. As a result, even if the refining and marketing company batches their order by purchasing every few days, it will not lead to inefficiencies in E&P companies.

Price fluctuations may cause the BWE as they often lead to forward buying by downstream actors (Lee et al., 1997a, 1997b). From the two companies investigated, it is found that forward buying is largely restricted by the storage capacity (i.e., the maximum volume of crude oil can be stored in the tanks or caverns) in the refining and marketing company.

In RM2, aside from the required inventory such as safety stocks, cycle stocks and volumes in transit, the company also has some inventory called "discretionary", which they can "choose to draw or fill based on market signals" as the manager said. Since the refining and marketing company usually has long-term leases on storage caverns or tanks, the storage cost is sunk. Over the recent five years, as the financing cost was about US$ 0.4 per month per barrel, the manager mentioned that the forward buying might be "motivated by inter-month price differences of less than US$ 0.50 per month". However, forward buying is restricted by the available storage capacity. As the manager of RM2 stated, "You may purchase early, but pre-bought volumes are limited by storage capacity. Typically, refiners do not have more than 20 days' worth of throughput (i.e., the maximum volume of crude oil can be processed in the refinery

per day) as storage capacity. On average, our tank storage capacity for crude oil is approximately 10–12 days' worth of throughput, and this has been generally stable during the past decade." Also, the refining and marketing company has to reserve some storage space in case that there is operational upset (i.e., an exceptional incident in operations such as an unexpected breakdown). This is because the refinery utilisation rate needs to be reduced (e.g., run with half or two-thirds of the throughput) during the operational upset to fix the issue, whereas the crude oil will not stop flowing in from the pipeline. Hence, enough room in the storage tank is needed to allow crude oil to accumulate during the operational upset. In the recent ten years, RM1 had only four to six days' worth of refinery throughput as crude oil storage capacity. Due to the storage capacity limitation, the company only keeps the inventory that is required, and a little extra so that when there is last-minute demand, they can take advantage of that opportunity if it is profitable. They do not do forward buying based on crude oil price like RM2. Hence, the impact of price fluctuation is alleviated by the limited storage capacity, thereby lowering the variation of monthly crude oil order quantity. In this case, we recognised that:

Finding 3. Limited storage capacity reduces crude oil order variability.

As expected, both managers stated that the crude oil supply is not limited as many sellers are available on the market. While they have a few major suppliers with whom they have ongoing relationships by signing long-term contracts or evergreen contracts (without an end but subject to termination upon notice), they never depend upon one. Instead, they deal with many suppliers and maintain options between supply locations. Therefore, rationing and shortage gaming does not exist in crude oil selling and purchasing.

As previously discussed, some shippers may over-nominate volumes to obtain adequate pipeline capacity from apportionment. However, both managers stated that they did not use such gaming strategy. First, gaming can be eliminated as a number of pipelines in North America allocate capacity based on historical shipping volume (e.g., a 12-month rolling average of the volume delivered). In this case, as the manager of RM1 mentioned, "Over-nominating doesn't help you. We'll only be allocated a space in line with what we've received historically." This is also why the pipeline capacity allocated to RM1 was relatively stable. Second, in the pipeline systems where the capacity is not allocated according to historical usage, there are rules around the nominations, which may reduce gaming to some extent. Specifically, the nomination verification process in Canada requires shippers to provide written third-party verification for supply availability and the capability to remove it from the delivery point. This means the company should have bought the amount of crude on the nomination before applying for pipeline capacity, and the maximum volume they can nominate is their throughput plus storage capacity subject to bottlenecks. Some pipeline companies also have the process to verify that the nominated volume will not be shipped through another pipeline, by rail or to another facility. Under these rules, over-nominating is largely restricted. Both managers mentioned that they carefully follow these rules to get their "fair share".

Nevertheless, shippers may still over-nominate within these rules. For example, in some pipeline systems, a share of capacity is contracted, while the rest is allocated based on total nominations. Suppose a company needs to ship 300 barrels and has contracted 150 barrels in the pipeline. Since there is no supply verification required for the 150 barrels contracted capacity, if the company expects a 30% apportionment, they may nominate 220 barrels to ensure they get the other 150 barrels of capacity and use the total 300 barrels supply as the verification for its nomination (220 barrels) without breaking any rules.

It should be noted that a company over-nominating in the pipeline system will not increase variations in their own crude oil order quantity. This is because the pipeline companies announce their apportionment percentage before the shipping period starts, and the refining and marketing company can return the volume of crude oil that cannot be shipped to the seller. Thus, over-nominating will not change the volume that can be delivered, which is the actual volume the refining and marketing company purchases. On the other hand, the apportionment percentage in a pipeline system fluctuates as the total nominations change from month to month. In 2018, among the four NEB-regulated pipelines in Canada, the apportionment percentages in three pipelines varied within a relatively small range (around 15%), while the other fluctuated from 0% to about 42% (NEB, 2018). Even though the apportionment percentage can be estimated with historical values, companies cannot predict it accurately, especially when there are large variations. As the percentage of apportionment influences the volume of crude oil delivered in each month, which is also the actual volume purchased by the refining and marketing company, fluctuations in the monthly apportionment percentages may increase variations in crude oil order quantity. This leads to the following proposition:

Proposition 2. *Variations in the percentage of pipeline apportionment increase the crude oil order variability*.

5.2. E&P company

O&G well drilling services ordered by the E&P company depends on its drilling activity level (i.e., the number of wells need to be drilled). The order quantity of drilling services is measured by rig operating days. In the existing literature, all the companies studied plan their production and determine their order quantity based on demand forecasting. However, for the two E&P companies investigated in this research, both managers stated that they do not consider demand forecast in drilling activity planning. As EP1's manager said, "It's not about how much we are able to sell but how much we are able to produce. Crude oil market is totally a liquid market, buying and selling are very easy. Crude producers would be able to sell however much crude they produce, and refineries will be able to buy however much they want if they can get it delivered". Also, the manager of EP2 remarked, "O&G is different from manufacturing. In manufacturing, they say 'we need to produce these many units in a year'; O&G is like 'we want to produce as many barrels of oil as possible'. We want to maximise our output, and we always think that every barrel of oil we produce will be able to sell." These are in line with Bradley's (2005) view that, companies producing commodities or commodity-like products, such as the E&P companies are supply-driven, because there are many buyers available in the market and demand can be assumed infinite. In contrast to the demand-driven company (e.g., manufacturer, wholesaler, retailer) whose objective is to supply predictable demand efficiently at lowest possible cost, supply-driven companies aim at maximising net profit, given full operating rates.

From the two cases, it is recognised that the crucial factor in drilling activity planning is the projected commodity price. Every year from August to November, market fundamentals people in the organisation make predictions for the average commodity prices in the next year. Based on the expected average O&G prices and the expected production rate, the company can estimate its cash flow, thereby determining the capital expenditure budget for the drilling activity in the next year. Besides, E&P companies usually have different areas (or global regions) to develop. Operational costs in a particular area depend on difficulties involved in extracting, processing and moving different types of products, as well as royalties, taxes, labour and material costs, etc. Thus, in different areas, the costs of getting one barrel of oil (or one unit of natural gas) ready to sell are different. Given the forecasts of commodity prices and operational costs, the company knows the most profitable products and the most profitable areas to develop. Hence, they can determine where to drill and how many wells to drill based on the available budget.

O&G prices fluctuate every day, while the drilling plan is updated at least on a quarterly basis when the E&P company observes a change in the general trend. When there is a general increasing trend in the

commodity price, the company may decide to drill more wells and order more drilling services as their cash flow from operations is projected to increase. "For example, one might plan to run five drilling rigs for a given calendar year, and a US$5 per barrel increase in the West Texas Intermediate (WTI) oil price could allow that company to pick up one more rig." the manager of EP2 mentioned. Though E&P companies try to be cautious in drilling activity planning (e.g., be conservative in commodity price forecasting, employ various scenarios and do sensitivity analysis), since their objective is to maximise net profit under full operating rates, during an oil boom, every company wants to increase production as soon as possible in order to generate more profit. As EP1's manager stated, "Basically in the boom, everybody wants the first oil as soon as possible, so that they can get that better market price for the product and economics would be better." When commodity price drops, the company's cash flow from operations will reduce. Hence, the company may lower the drilling activity level by cancelling a number of drilling programs or even stop drilling totally but sell the existing production to cover the overhead costs. In the two cases investigated, both managers mentioned that they and many other companies had largely cut down their drilling programs during the recent 2014–2015 oil crash. However, as they had expected a high drilling activity level previously during the oil boom, many drilling rigs were under long-term contracts. They thereby paid a large amount of penalties to offset contractual obligations when the unpredictable oil crash occurred.

In contrast, E&P companies seldom cut down their production rates when the commodity price goes down. As the manager of EP2 explained, this is because "Upstream oil production is largely a 'sunk cost' business. Exploration and capital cost of drilling making up a large percentage of the total cost of production. Marginal cost of production is almost always lower than the marginal sales value. And even when the marginal sales value gets close to the marginal cost, there is a danger of reducing production volume as unit cost competitiveness will diminish and economies of scale will be lost." EP1's manager also mentioned that they do not reduce production when price goes down: "We do not reduce production. We do that based on pipeline spill, explosion, so they can do repairs and figure out what the root cause was. Except for that situation, we will not reduce production just because for example the price. We try to wait for the price to come back." As a result, the quarterly drilling service order quantity (i.e., the rig operating days contracted per quarter) fluctuates more than the quarterly crude oil production quantity. Hence, commodity price fluctuation is a major cause of BWE in the E&P company. We thus present the following proposition:

Proposition 3. *Commodity price fluctuations increase variability in orders for drilling services*.

In addition to commodity prices, many other factors are also considered in drilling activity planning, which impact the variations in orders for drilling services.

According to the two managers interviewed, some E&P companies operate their drilling programs all year-round. Hence, the drilling rig fleet contracted generally remains stable and flat. However, some choose to perform the majority of their projects in the first quarter (e.g., when the company has a small number of wells to drill), so that they can turn on production and generate the most profit in the rest of the year. For some areas, the rainy weather in spring and summer, farming activities and/or the protection of valuable animals influence the movement of drilling equipment, so the company cannot drill all year long. For example, due to the rainy weather in spring and summer, most of the drilling activities in Canada are done from January to March, ramping up again through July to December. According to the industry-specific database CanOils, over the past 15 years in Canada, the ratio between the number of wells drilled by an E&P company in the first quarter and that in the second quarter was about five on average, and could be as high as over 15. Besides, this ratio varied among different companies and changed from year to year. Thus, the rig operating days contracted are higher in some quarters than in others, which results in variations in quarterly drilling service order quantity. As a result, we found:

Finding 4. Variability in orders for drilling services can increase as the E&P company intentionally aggregates its drilling activities to maximise profit or due to seasonality.

After the E&P company has signed the contract to lease a piece of land from the government or a private owner, the company needs to drill within a certain time. Otherwise, the land right will be returned to the owner. Thus, time limitation in the lease forces the company to drill a number of wells within a certain timeframe.

E&P companies also have commitments which influence the drilling activity level. Specifically, if the company has a joint venture partner in developing a property, they need to drill a number of wells to reach a certain production rate so that they can guarantee the return rate required by the partner. Further, E&P companies usually have commitments to pipeline companies as they contract pipeline capacity for crude oil shipping. Some companies may also have agreements with rail. Since the well production rate in the reservoir declines with cumulative production, such commitment also encourages the E&P company to drill a number of wells to keep the production rate consistent, so that the assigned capacity is fully utilised and the prepaid money is not wasted. For example, from 2016 to now, EP2 had transportation commitment of approximately 190,000 bbls/d.[2] Also, in both EP1 and EP2, their production rate has not been lower than their transportation commitment in the past decade. However, it should be noted that when there is a substantial or extended decline in commodity prices, some companies may have unutilized transportation commitment.

Additionally, the total number of wells that can be drilled in a section of land, the capacity of processing facility and the capacity of pipeline all limit land development. For instance, government regulations limit the number of wells on a piece of land to avoid excessive drilling and reduce the risk of fire or blowout (Harrison, 1970). In Canada, the normal surface well spacing in Alberta is one well per section of land (640 acres) for gas and one well per quarter section of land (160 acres) for oil (AER, 2011). Regarding the processing facility, the manager of EP1 mentioned that one of their crude oil processing facility has a capacity of 72,000 bbls/d, which has not been expanded in the past decade. However, the processing capacity constraint can be relaxed by expansion when the company has sufficient budget. At the time of data collection, in two of EP2's reserves, they had a total oil processing capacity of about 390,000 bbls/d and was expecting to expand the capacity by 30,000 to 50,000 bbls/d in the future. Besides, pipeline capacity limits the growth of production, thereby impacting the well drilling activity in an E&P company.

As the drilling activity level is constrained within a certain range by the factors discussed above, variation of quarterly drilling service order quantity is reduced. Accordingly, we propose that:

Proposition 4. *Time limitation in the land lease, commitments to the joint venture partner and pipeline company, number of wells that can be drilled, processing facility capacity limitation and pipeline capacity limitation all reduce the variability in orders for drilling services*.

In supply chain management, the well-known risk-pooling effect suggests that demand variability is reduced if we aggregate demand across locations, since it is likely that high demand from one customer will be offset by low demand from another (Simchi-Levi et al., 2008). Sucky (2009) showed that the BWE might be overestimated when a supply chain is assumed in the analysis, ignoring the risk pooling effects that can be utilised in the supply network. From the perspective of marketing, Bishop et al. (1984) argued that in oil- and gas-related turbomachinery manufacturing, diversity of operations, in markets and product lines can reduce variability in demand.

Though E&P companies are supply-driven, they often have different

[2] Barrels per day.

types of commodity assets (e.g., light oil, heavy oil, natural gas) and different development areas (or global regions). As the manager of EP2 mentioned, "If your company has both heavy and light oil assets, when the commodity price of a barrel of oil is low, you may stop drilling heavy oil wells, and you might drill more wells in the light oil assets that are cheaper to produce. When the natural gas price is very strong while crude oil price is low, we choose to pursue rocks saturated with lots of natural gas; when the natural gas price is low, we drill more wells with oil." Hence, when commodity prices fluctuate, the drilling activity level of an E&P company with various types of assets would be more stable.

Crude oil prices are different in the global market, North American market, and Western Canadian market, since the supply-demand balances differ in different markets. Specifically, Brent crude is traded on the global market; WTI is the benchmark of American crude. Due to the growth of American crude production in the past a few years, WTI is priced at a discount to Brent crude, as there is more supply than demand within the inland U.S. than in the global market. The same concept applies to Western Canadian crude, such as the Edmonton Light. Edmonton Light is of the same quality as WTI but is less expensive. This is because there is more crude production in western Canada than there is available pipeline capacity to move it. Hence, the producer has to discount it to stimulate the customer to buy Edmonton Light. Therefore, if a company has assets in different regions, when the asset in western Canada is not profitable due to the low oil price, the company can choose to drill wells in the U.S. or other global regions. Thus, the drilling activity level will be more stable than those with assets only in western Canada.

According to the manager of EP1, "Now, a lot of companies are very much on the quarter by quarter basis, but not our parent company. Our parent company is more conservative and takes a longer view. So, they are gonna wait until they see a demonstration that Alberta could attain narrowing that differential between Western Canadian crude and WTI before they can make a lot of investment here again. Otherwise, they can be profitable by investing in the Middle East, Asia, South America etc. In the past two years, we did not do any drilling here. If we were able to achieve WTI or Brent price, we would have way more money being spent in drilling." Therefore, the diversification in types and locations of the assets lowers the E&P company's exposure to commodity price risk, thereby reducing the fluctuations in drilling activity level. As a result, the quarterly drilling service order quantity is less variable. Hence, we bring up the following proposition:

Proposition 5. *Asset diversification reduces the variability in orders for drilling services.*

BWE is considered an outcome of the lack of coordination in the supply chain (Chopra and Meindl, 2010). Lee et al. (1997b) proposed that sharing end-user demand information can mitigate the BWE generated by demand forecast updating, which has then become the most commonly investigated coordination mechanism (e.g., Chatfield et al., 2004; Dejonckheere et al., 2004; Hussain and Drake, 2011; Ma et al., 2013). In regard to vertical integration, Osegowitsch and Madhok (2003) showed that integrating downstream part in the supply chain greatly benefits the manufacturer as it gives the manufacturer more accurate and timely information of customers' demand, inventory, and requirements for products and services. This result was supported by Guan and Rehme (2012) who demonstrated that vertical integration could eliminate the boundaries between upstream and downstream supply chain members, thereby improving supply chain visibility and giving companies access to detailed information about their successive sections in the supply chain. Therefore, vertical integration should be able to mitigate the BWE through information sharing.

However, as E&P companies are supply-driven, which does not consider demand forecasting in drilling activity planning, downstream information is not shared in the integrated O&G company either. EP2's manager said that they do not need the demand information from their downstream refineries. The manager of RM2 mentioned that they only share information of pipeline transportation and refinery maintenance with their upstream section. Moreover, as crude oil is sold on the commodity market with numerous sellers and buyers available, EP2's manager stated, "Our refinery may not purchase crude oil from its upstream. The crudes usually are sold to many different companies, and refinery can purchase crude oil from different companies as well." Similarly, the manager of RM2 said, "Upstream production does not always feed the closest refinery. Both upstream and downstream search for the best offer in the market. We often sell our production to others, buy refinery supply from others, sell our refined products to others and supply our gas stations from others. […] In the financial report, our balance is approximately 1 to 1 as it relates to amount produced versus amount consumed, but the actual amount of our own crude as feedstock is about 40% during the recent five years, as we find quality, location, and transportation advantages to selling and buying crudes. The proportion of crude used as its own feedstock is typically low in integrated companies. NOCs may be an exception." Therefore, managers of EP2 and RM2 both agreed that in integrated O&G companies, upstream and downstream are generally independent of each other in production planning.

While the vertical integration does not facilitate information sharing between upstream and downstream in the O&G company, business diversification resulting from the vertical integration impacts the company's cash flow, which is crucial in drilling activity planning. In both integrated companies studied in this research, total corporate cash flow expectation is considered in determining the budget for drilling activities. The two managers agreed that involvement in downstream refining could generate more steady cash flow, thereby reducing the variation of drilling activity level to some extent. The manager of RM2 said, "Refining revenues tend to be more regular than the upstream. Location discounts, especially here in Canada (i.e., as aforementioned, Western Canadian is priced at a discount to WTI), can hurt pure producers to the advantage of refiners. So, we have chosen to have exposure to refining revenues, so that we can offset the risk of location discounts in Canada. This should reduce the fluctuations in the expenditure spends on upstream drilling." EP2 has a similar statement on its website, which says their ownership in the refineries allows them to capture value from crude oil production to output of finished products, thereby reducing the risk of commodity price fluctuations. Thus, vertical integration towards downstream lowers the company's exposure to commodity price risk, generates more stable cash flow, and reduces the fluctuations in drilling activity level. Accordingly, variation of quarterly drilling service order quantity is reduced. Therefore, the following proposition is presented:

Proposition 6. *Vertical integration towards downstream reduces variability in orders for drilling services.*

Specifically, the level of vertical integration refers to the parts of the value network that belong to the company (Saccani et al., 2007). A company's level of vertical integration increases as more value chain activities are performed internally.

5.3. Drilling contractor

Supplies and components ordered by the drilling company can be divided into two categories: consumable items and capital items. Consumable items are used for rig operation and maintenance, which include fuel, lubricants and some small parts of the drilling rig that have a short lifespan and need to be replaced frequently (e.g., shaker screens, pump parts, gaskets, seals). Capital items consist of large rig components and equipment replacement parts. Large rig components are ordered for maintenance, upgrading and building new drilling rigs, which include engines, top drives, mud pumps, generators, draw works, etc. Within the large rig components, there are also some major parts such as the replacement modules for the fluid end of mud pump, transmission on an engine, etc. These equipment replacement parts are ordered for maintenance and upgrading.

Consumable items are similar to raw materials used for production in manufacturing companies, whereas the capital items are replaced on a depreciation basis. Hence, factors that influence the order variability are analysed separately for the two types of items.

For the industries studied previously, demand faced by the retailer, wholesaler and manufacturer are stochastic. However, according to the two managers from the drilling company, demand for consumables items used in well drilling is generally known in advance with very low uncertainty. This is because based on the type of drilling rig used and the design of the project, they can accurately predict the demand for consumable items based on historical data. Consumable items are also used for scheduled preventative maintenance on the working drilling rigs. This part of demand can be precisely estimated as well. Besides, the consumable items are ordered weekly. The lead time of consumable items is only a few days, which is short and stable. For example, the manager of DC1 mentioned that they usually place orders on Sunday, and the consumable items can be received on Wednesday. Also, there is generally no special discounts or promotions on consumable items incentivising forward buying by the drilling company. In addition, as the supply of consumable items is sufficient, the problem of rationing and shortage gaming does not exist in orders of consumable items. Therefore, variations in the weekly order quantity of consumable items are unlikely to be greatly larger than that in the demand quantity. In this case, severe BWE should hardly exist in consumable items, since the only potential cause of BWE in consumable items is that their orders are slightly batched on a weekly basis. This leads to the following proposition:

Proposition 7. *Substantial BWE is unlikely to exist in orders of consumable items used in O&G well drilling.*

Similar to the turbomachinery equipment studied by Bishop et al. (1984) and the machine tool case investigated by Anderson et al. (2000), orders of large rig components and equipment replacement parts consist of two parts: replacement orders for maintenance and non-replacement orders generated from new demand such as upgrading and new drilling rig building.

For large rig components and equipment replacement parts, orders of replacement depend on their lifetime. The life expectancy varies among components and parts. As the two managers mentioned, the drilling company uses the parts and components on idle drilling rigs or keeps a few units in stock as backups. Drilling rigs are inspected during the scheduled preventative maintenance. If a piece of equipment appears to be worn out or about to wear out upon inspection, it will be replaced. Large rig components are also changed around the end of lifecycle recommended by the OEM. Once an equipment replacement part or large rig component is substituted by the spare in stock or on the idle drilling rig, a new one will be ordered immediately. In this case, large rig components and equipment replacement parts are replaced at a relatively stable rate based on their expected lifespan. However, it should be noted that the replacement demand may fluctuate in practice. As the manager of DC2 emphasised, "The lifecycle of a component depends upon maintenance. Just like a car, maintained equipment will last longer than abused equipment." Therefore, the replacement rate may not necessarily hold constant. Also, the drilling rigs are not built up in the same year. For newly built drilling rigs, the parts and components do not need to be replaced in the short run. In addition, the two managers also mentioned that unexpected breakdown might happen. Though uncommon, it will influence the replacement order as well. In general, a certain number of equipment replacement parts and large rig components will be replaced each year. Although the replacement orders have some variations, both managers agreed that there are no large fluctuations in the annual (or quarterly) replacement order quantity.

Significant order variability of large rig components and equipment replacement parts is caused by non-replacement orders generated from new drilling rig building and drilling rig upgrading. Every year, drilling companies predict their total rig operating days and cash flow in the coming year, thereby determining the capital expenditure budget spending on equipment replacement parts and large rig components used for upgrading and rig fleet expansion. Similar to the E&P company, which updates its drilling plan at least on a quarterly basis when the general trend of commodity price changes, drilling contractors also adjust their capital expenditure budget quarterly or semi-annually.

When the drilling company predicts future rig operating days, some of their rigs may have already obtained signed contracts. According to managers from DC1 and DC2, drilling rigs are contracted through fixed-term or open contracts. Fixed-term contracts often range from six to 24 months, but can be as long as five years. They specify the number of wells to drill and allow termination by the customer (i.e., E&P company). If the contract is terminated before it expires, depending on the clause in the contract, the E&P company may need to pay a shortfall rate for the remaining period, contract another rig to use up the remaining days, or transfer the unused days to another contract. In the open contract, the E&P company only states, but does not guarantee the number of wells required to be drilled. For example, if the drilling rig does not perform well or the commodity price changes, the E&P company may terminate the contract without paying any penalty. Thus, when forecasting operating days for rigs under signed contracts, the drilling company adds together the guaranteed days from fixed-term contracts and estimated days from open contracts. Particularly, the estimation from open contracts is based on customers' approved budget. Also, the drilling contractor assumes that they have the capability to satisfy the customer and complete the entire drilling program. For drilling rigs not under contracts, the operating days are estimated based on historical operations, commodity price forecasts, market sentiment, customers' verbal commitments, and the manager's judgemental estimation of the likelihood that they can get work from a customer. Thus, forecasting by a drilling company can be more complex and involve many more factors than with the time series methods often assumed in the BWE literature.

The cost of building a new drilling rig has increased markedly in recent years as E&P companies require high technology and more efficient drilling rigs for unconventional drilling programs. The manager of DC1 mentioned, "About 15 years ago, the company might build up a rig for US$4 million, but now, in recent five years, the average cost would be US$15 million, an expensive one would be over US$20 million." Also, the large rig components used in drilling rig building has a long lead time (usually six months to one year) and can be as long as a year and a half during busy time. On the other hand, the commodity price is highly volatile and difficult to predict for the long term. In this case, long fixed-term contract (two to five years) are generally required to be signed with the E&P company before the drilling contractor orders the large rig components used for new drilling rig building. During an oil boom, large drilling contractors such as DC2 may allocate some capital budget to build a new rig without a signed contract. However, when they are building the rig, they look for a contract. Thus, the rig can get contracted as soon as it is built. The company will not start building the next rig until they have secured a contract for the first rig being built. This process is carried on to the subsequent rigs, which thereby only exposes the company to potentially having the last rig being built not under contract. The manager of DC2 said that they built a number of drilling rigs in 2014 when the crude oil price sustained high levels and all the rigs got contracted when they were built up. From their annual financial report, the capital expenditure in 2014 increased by 53% compared to 2013. As the drilling rig is built in one year which dramatically increases the order quantity of large rig components, while the service demand for that rig is distributed in several years under the term contract, variation of annual order quantity of large rig components is larger than that of the drilling service demand. Thus, the BWE is incurred through the multiplier effect explained earlier.

For drilling rig upgrading, both managers stated that if the upgrading is required by the customer, they will always negotiate for a fixed-term contract. Drilling rigs may also be upgraded without contracts signed when there is sufficient budget, the drilling contractor wants to stay

competitive, the upgraded drilling rig is very likely to get contracted, and the day rate is expected to increase. So, the company can cover the cost and become more profitable. However, as no one is certain about the commodity price in long-term and oil price crash is generally unpredictable, such upgrading may result in more severe BWE through the multiplier effect, if the demand does not increase as expected. In the case of DC1, the manager said, "In 2014, we upgraded multiple rigs with new pumps and hydraulic catwalks to keep up with market demand. But, the price of oil collapsed at the start of 2015 and the demand for rigs dropped rapidly. We were left with upgraded pumps and hydraulic catwalks on our rigs that did not get used. Probably spent around US$2.2 million." Therefore, it appears that the multiplier effect in orders of large rig components and equipment replacement parts is the primary cause of BWE in drilling companies.

6. Discussion

6.1. BWE factors in the O&G supply chain

This multiple-case study identified a number of factors that cause or smooth the BWE in the O&G supply chain. Regarding refining and marketing companies, they are similar to the companies studied in the existing literature, which determine their crude oil order quantity based on demand forecasting. However, since the long-term contracts used in refined products sales significantly increase the certainty in future demand, and the additional information considered in demand forecasting provides explanation for the temporary change in historical demand data, the problem of demand forecast updating is mitigated in the refining and marketing company and the impact of long lead time in crude oil supply is attenuated as well. In the existing studies, some researchers argued that production capacity limitation is a potential cause of the BWE (De Souza et al., 2000; Lee et al., 1997b), but other facilities (e.g., storage, transportation) in the supply chain are generally regarded as uncapacitated. Nevertheless, through our investigation, it is found that the capacity limitations of transportation and storage play important roles in the refining and marketing company. Due to the limited and stable pipeline capacity, demand forecasting may no longer be decisive in determining crude oil order quantity. Restrictions on storage capacity can eliminate the forward buying caused by price fluctuations.

Though the refining and marketing company reviews its crude oil inventory and determines the order quantity on a monthly basis similar to many manufacturing companies, as crude oil is traded on the commodity market with a highly volatile price, the company usually orders a certain volume of crude oil every day or every few days to reduce its exposure to price risk. Hence, order batching is not common in crude oil purchasing. The commodity market also eliminates the rationing and shortage gaming between the E&P company and refining and marketing company, as numerous sellers and buyers are available.

In the refining and marketing company, the factor discovered that might increase the BWE is the fluctuations in pipeline apportionment. When the pipeline is oversubscribed, gaming within the shipping nomination rules is possible. Nonetheless, gaming itself will not increase the crude oil order variability in the company, because the extra volume can be returned to the seller before the shipping starts. Instead, the volume of crude oil delivered, which is also the actual quantity of crude oil ordered, is directly impacted by the pipeline apportionment percentage. As the apportionment percentage cannot be accurately predicted, variations in the percentage may amplify crude oil order variability in the refining and marketing company.

In summary, except the pipeline apportionment percentage fluctuations, most factors examined in the refining and marketing company appear to reduce rather than increase the BWE. These findings may provide potential explanations to the result in Bray and Mendelson (2012) that there was no severe BWE in petroleum and coal manufacturing companies, which include the refining and marketing company.

E&P companies are distinct from the companies studied previously in two aspects. First, E&P companies are supply-driven instead of demand-driven, whose objective is to maximise their net profit under full production rates. Therefore, demand forecasting is not the basis for drilling activity planning in E&P companies. Lead time thereby cannot aggravate the BWE through demand forecast updating. Second, the drilling service ordered by the E&P company is intangible, heterogeneous, and cannot be stored and transported since production and consumption occur simultaneously (Akkermans and Vos, 2003). In this case, it is not surprising that other major causes of the BWE proposed by Lee et al. (1997a, 1997b), i.e., order batching, forward buying resulting from drilling service price fluctuations and shortage gaming were also not found in the two E&P companies investigated in this study.

Particularly, the major potential cause of BWE found in the E&P company is commodity price fluctuation. The reason behind is that the commodity price largely impacts the cash flow in the E&P company, which thereby influences the budget for drilling activities. During an oil boom, as E&P companies generate more cash flows from their operations and have a larger budget for drilling, they drill more wells to increase their production so as to gain more profit. When the oil crash occurs, they slow down or stop drilling since the budget decreases. However, they seldom cut down their production rate when commodity price drops. Thus, orders for drilling services are more variable than O&G production. Besides, variations in orders for drilling services may increase due to seasonality or when the E&P company intentionally aggregates its drilling activities at the beginning of the year so as to start production earlier and generate more profit.

On the other hand, we found that asset diversification might smooth variability in orders for drilling services, since assets with various types and locations can reduce the company's exposure to commodity price risk, leading to more stable drilling activity level.

More importantly, our finding in the integrated O&G company challenges the prevailing theory, which suggests that vertical integration, as an extreme of vertical coordination of the supply chain (Hobbs and Young, 2000), can facilitate information sharing between upstream and downstream sections by eliminating the boundaries, hence mitigate the BWE. In contrast, due to the availability of numerous sellers and buyers in the commodity market, the upstream section in the integrated O&G company is usually not the major supplier for its downstream section, as both parts seek best offers in the market to sell and purchase crude oil. Also, the supply-driven E&P section in the integrated O&G company does not need the demand information from downstream to plan its production and drilling activities. As a result, vertical integration does not reduce variability in drilling service orders through information sharing. Instead, integrating downstream sections can decrease the company's exposure to commodity price risk and generate more steady cash flows and budget for drilling activities. Consequently, vertical integration towards the downstream may reduce the variations in drilling service order quantities. From the above discussion about the factors that trigger or smooth the BWE in E&P companies, cash flow can be considered as an important mediator in the relationship between commodity price fluctuations and the BWE as well as between vertical integration and the BWE, which is rarely mentioned in the existing literature.

In addition, variability in orders for drilling services is also constrained by time limitations in land development, commitments to joint venture partners and pipeline companies, the total number of wells that can be drilled, as well as capacity limitations in the processing facility

and pipeline. These are unique factors found in E&P companies that smooth the BWE.

O&G well drilling service differs from the labour-intensive services such as banking, insurance or the telecom service studied by Akkermans and Vos (2003) and Akkermans and Voss (2013), in that it requires large drilling equipment and various supplies in drilling rig operation. Among the products ordered by the drilling contractor, consumable items used in drilling rig operation and regular maintenance are similar to the raw materials used in manufacturing. Our case study found that orders of consumable items are unlikely to be significantly more volatile than the demand. This is mainly because the demand for consumable items is not stochastic but largely known with certainty for a particular project and drilling rig based on historical usage. Also, except that the ordering of consumables items is batched on a weekly basis, no other potential cause of the BWE is found. For drilling contractors, it appears that orders of large rig components and equipment replacement parts are the main source of BWE. Similar to the capital equipment studied in the existing studies, a number of large rig components and equipment replacement parts that reach their lifespan are replaced every year, whereas the demand generated from drilling rig building and updating can lead to substantial order variability through the multiplier effect. Additionally, drilling rig upgrading without a fixed-term contract signed in advance may aggravate the BWE if commodity price drops and the demand for drilling service does not increase as expected.

6.2. Implications for related industries

Some propositions and findings may apply not only to O&G but also to other industries. Commodity price fluctuations, which is a major cause of the BWE in E&P and upstream sections in integrated O&G companies, could also drive the BWE in mining companies of other resource extraction industries. This is because producers of metals and minerals (e.g., coal, iron, copper, gold) are subject to market prices that are volatile and difficult to forecast as well, since their products are also commodities (Cameron and Stanley, 2017). Similar to E&P companies, large mining companies have high capital investments in exploration and drilling activities (Cameron and Stanley, 2017; Zuñiga et al., 2015). Thus, commodity price fluctuations may also increase the BWE in mining companies, as it impacts the firm cash flow, which is a critical factor in determining the capital expenditure on exploration and drilling. The firm cash flow may also work as a mediator in the relationship between the commodity price fluctuations and the BWE in mining companies. Furthermore, vertically integrated companies (i.e., from mining to refinery) with diversified assets, may have reduced BWEs due to lower commodity price risk and more stable cash flow. Nevertheless, different from companies in the O&G industry which are exclusively capital intensive, there are many artisanal and small-scale companies in the mining industry that are labour intensive with very little capital investment (Cameron and Stanley, 2017). Hence, it should be noted that the propositions and findings discussed above might not be applicable to these companies.

Some results may apply to an even more general context. First, long-term sales contracts with definite delivery quantities could always smooth a supplier's BWE by reducing demand uncertainty. Second, demand forecasting methods properly utilising additional information that explains demand fluctuations, could smooth the impact of forecast updates, thereby reducing variations in supply orders. In previous BWE research, the demand forecasting approaches commonly discussed are time series methods, whereas in practice, forecasting techniques applied can vary. Besides the basic time series methods and advanced statistical models, some companies use statistical software to generate initial predictions, then adjust the forecasting results by expert's judgement considering other factors and information that are difficult to be included in a statistical model (Trapero et al., 2011). While poor judgmental adjustments could introduce biases and undermine the forecasting accuracy (Fildes et al., 2006; Trapero et al., 2011), with rich information and suitable adjustments based on essential domain knowledge, such integration of quantitative forecast and managerial judgment can lead to more accurate forecasting results (Fildes et al., 2006), thereby alleviating the BWE caused by demand forecast updating. For instance, taking the effect of retail promotion campaign into account, the manager can have a better knowledge of the increase in sales and adjust the proportion of historical data fed forward into the future, so as to reduce the likelihood of overordering and excessive inventory. Finally, supply chains with localised storage constraints could curb forward buying behaviour, leading to reductions in BWE.

7. Conclusion

Our findings contribute to on-going research on the BWE, which thus far has primarily delved into manufacturing, wholesale and retail. In resource extraction industries, while the existence of BWE in the O&G industry has been demonstrated in several empirical studies (e.g., Bray and Mendelson, 2012; Cachon et al., 2007; Isaksson and Seifert, 2016; Shan et al., 2014), the causes are unexplored. Also, whether the major causes of BWE summarised in the prevailing literature are applicable to the O&G industry is unclear, because it has several features that are distinct from the industries studied previously. Our research filled these gaps by investigating different types of companies in three main echelons of the O&G supply chain. Findings in this paper showed that the existing theories of BWE could offer explanations for the phenomenon in the refining and marketing company and drilling company. However, the pipeline transportation system plays a rather special role in crude oil procurement. On the one hand, pipeline apportionment percentage fluctuations amplify the BWE in refining and marketing companies; on the other, limited and stable pipeline capacity may largely alleviate the problem. Besides, it appears that substantial BWE may not exist in all types of products ordered by the drilling company, as the factor that may cause significant BWE was only found in the orders of capital items.

Theories in the prevailing literature have limitations, which cannot provide explanations for the BWE in the E&P company. Instead, unique factors drive or mitigate the BWE such as the commodity price fluctuations, asset diversification and vertical integration were identified. Besides, in the mechanisms of how the commodity price fluctuations and vertical integration influence the BWE, cash flow plays as an important mediator, which was barely discussed in the existing studies. Furthermore, downstream demand information sharing, a widely advocated countermeasure of BWE, does not appear relevant in the integrated O&G company due to the commodity market and the supply-driven E&P section. We believe that these findings would provide valuable insights for the application and limitation of existing BWE theories and enable better understanding of this phenomenon in the upstream end of the supply chain. Moreover, some propositions and findings can be generalised to other resource extraction industries, and in some cases, even to traditional manufacturing, wholesale and retail supply chains. The distinct factors mitigating the BWE discovered in the study, such as asset diversification and vertical integration towards downstream may guide the enterprises in finding potential countermeasures for the BWE in practice.

One limitation of this multiple-case study is the number of sampled cases. Eisenhardt (1989) suggested that a number of cases between four and ten is desirable for theory building. As we only included six cases with two cases for each echelon in the supply chain, it may limit generalizability. Another limitation in case selection is that our case study focuses on the companies in North America. Also, no IOC, NOC, nor any offshore drilling company was included in our cases. However, it should be noted that for the two E&P companies studied, EP2's manager stated that they have business units in different geographical regions, which have to propose their drilling plans to get budget approval from the headquarters. The manager of EP1 said, "I cannot say we are probably too much different than Exxon Mobil as an example, as we got multiple country activities." Besides, through the communication with

the manager of EP1 and examining the annual report of Transocean (one of the largest offshore drilling companies in the world), it appears that onshore and offshore drilling have many similarities. For instance, offshore drilling also requires both consumable items and capital items. Demand for capital items in offshore drilling is generated from replacement, drilling rig building and upgrading as well. As with onshore drilling, there are also contracted and non-contracted drilling rig building and upgrading in the offshore drilling company. Hence, we believe that the findings in this study can also shed some light on the understanding of BWE in IOCs and offshore drilling contractors. Nonetheless, differences can still exist, especially in NOCs that involve in the entire O&G supply chain from refining and marketing to equipment manufacturing such as PetroChina. Therefore, a direction for future research could be focusing on validating the findings with a higher number of cases, in other global regions and in IOCs, NOCs and offshore drilling companies.

Lastly, as this study follows a qualitative multiple-case study design, we recommend performing quantitative studies based on this research. For example, public databases and survey can be used to collect data from a larger sample of companies, so that the propositions put forward can be tested in a broader context. Studies based on simulation and mathematical programming may also be considered to validate our findings. Finally, the results can probably be replicated in other industries. Although the propositions on fluctuations in pipeline apportionment and internal constraints (e.g., commitments to transportation companies, processing facility capacity limitation) may be valid only in O&G, others relating to commodity price fluctuations, asset diversification and vertical integration could also apply to related mining industries.

Declaration of competing interest

None.

Acknowledgements

This research was partially funded by the Warren Dyer Fellowship and CN Rail through the Canadian Centre for Advanced Supply Chain Management and Logistics at the Haskayne School of Business, University of Calgary.

References

AER, 2011. AER Brochure: Understanding Oil and Gas Development in Alberta. Alberta Energy Regulator, Alberta, Canada.
Akkermans, H., Vos, B., 2003. Amplification in service supply chains: an exploratory case study from the telecom industry. Prod. Oper. Manag. 12, 204–223. https://doi.org/10.1111/j.1937-5956.2003.tb00501.x.
Akkermans, H., Voss, C., 2013. The service bullwhip effect. Int. J. Oper. Prod. Manag. 33, 765–788. https://doi.org/10.1108/IJOPM-10-2012-0402.
Ancarani, A., Di Mauro, C., D'Urso, D., 2013. A human experiment on inventory decisions under supply uncertainty. Int. J. Prod. Econ. 142, 61–73. https://doi.org/10.1016/j.ijpe.2012.09.001.
Anderson, E.G., Fine, C.H., 1999. Business cycles and productivity in capital equipment supply chains. In: Tayur, S., Ganeshan, R., Magazine, M. (Eds.), Quantitative Models for Supply Chain Management. Springer US, Boston, MA. https://doi.org/10.1007/978-1-4615-4949-9_13.
Anderson, E.G., Fine, C.H., Parker, G.G., 2000. Upstream volatility in the supply chain: the machine tool industry as A case study. Prod. Oper. Manag. 9, 239–261. https://doi.org/10.1111/j.1937-5956.2000.tb00136.x.
Behar, M.A., Ritz, R.A., 2016. An Analysis of OPEC's Strategic Actions, US Shale Growth and the 2014 Oil Price Crash. International Monetary Fund.
Bhattacharya, R., Bandyopadhyay, S., 2011. A review of the causes of bullwhip effect in a supply chain. Int. J. Adv. Manuf. Technol. 54, 1245–1261. https://doi.org/10.1007/s00170-010-2987-6.
Bishop, W.S., Graham, J.L., Jones, M.H., 1984. Volatility of derived demand in industrial markets and its management implications. J. Mark. 48, 95–103. https://doi.org/10.2307/1251514.
BP, 2018. BP Statistical Review of World Energy 2018 (London, UK).
Bradley, Z.H., 2005. Are supply (driven) chains forgotten? Int. J. Logist. Manag. 16, 218–236. https://doi.org/10.1108/09574090510634520.
Bray, R.L., Mendelson, H., 2012. Information transmission and the bullwhip effect: an empirical investigation. Manag. Sci. 58, 860–875. https://doi.org/10.1287/mnsc.1110.1467;1910.1287/mnsc.1110.1467.
Cachon, G.P., Randall, T., Schmidt, G.M., 2007. In search of the bullwhip effect. Manuf. Serv. Oper. Manag. 9, 457–479. https://doi.org/10.1287/msom.1060.0149.
Cameron, P.D., Stanley, M.C., 2017. Oil, Gas, and Mining: A Sourcebook for Understanding the Extractive Industries. The World Bank.
Chatfield, D.C., Kim, J.G., Harrison, T.P., Hayya, J.C., 2004. The bullwhip effect-impact of stochastic lead time, information quality, and information sharing: a simulation study. Prod. Oper. Manag. 13, 340–353. https://doi.org/10.1111/j.1937-5956.2004.tb00222.x.
Chen, F., Drezner, Z., Ryan, J.K., Simchi-Levi, D., 2000. Quantifying the bullwhip effect in a simple supply chain: the impact of forecasting, lead times, and information. Manag. Sci. 46, 436. https://doi.org/10.1287/mnsc.46.3.436.12069.
Chopra, S., Meindl, P., 2010. Supply Chain Management: Strategy, Planning, and Operation. Prentice Hall.
Croson, R., Donohue, K., 2006. Behavioral causes of the bullwhip effect and the observed value of inventory information. Manag. Sci. 52, 323–336. https://doi.org/10.1287/mnsc.1050.0436.
De Souza, R., Zice, S., Chaoyang, L., 2000. Supply chain dynamics and optimization. Integr. Manuf. Syst. 11, 348–364. https://doi.org/10.1108/09576060010335627.
Dejonckheere, J., Disney, S.M., Lambrecht, M.R., Towill, D.R., 2003. Measuring and avoiding the bullwhip effect: a control theoretic approach. Eur. J. Oper. Res. 147, 567–590. https://doi.org/10.1016/s0377-2217(02)00369-7.
Dejonckheere, J., Disney, S.M., Lambrecht, M.R., Towill, D.R., 2004. The impact of information enrichment on the Bullwhip effect in supply chains: a control engineering perspective. Eur. J. Oper. Res. 153, 727–750. https://doi.org/10.1016/s0377-2217(02)00808-1.
Disney, S.M., Towill, D.R., 2003. Vendor-managed inventory and bullwhip reduction in a two-level supply chain. Int. J. Oper. Prod. Manag. 23, 625. https://doi.org/10.1108/01443570310476654.
Dubois, A., Araujo, L., 2007. Case research in purchasing and supply management: opportunities and challenges. J. Purch. Supply Manag. 13, 170–181. https://doi.org/10.1016/j.pursup.2007.09.002.
Duc, T.T.H., Luong, H.T., Kim, Y.-D., 2008. A measure of the bullwhip effect in supply chains with stochastic lead time. Int. J. Adv. Manuf. Technol. 38, 1201–1212. https://doi.org/10.1007/s00170-007-1170-1.
Eisenhardt, K.M., 1989. Building theories from case study research. Acad. Manag. Rev. 14, 532–550. https://doi.org/10.2307/258557.
Eisenhardt, K.M., Graebner, M.E., 2007. Theory building from cases: opportunities and challenges. Academy of Management Journal 50, 25–32. https://doi.org/10.5465/AMJ.2007.24160888.
Fildes, R., Goodwin, P., Lawrence, M., 2006. The design features of forecasting support systems and their effectiveness. Decis. Support Syst. 42, 351–361. https://doi.org/10.1016/j.dss.2005.01.003.
Forrester, J.W., 1958. Industrial dynamics: a major breakthrough for decision makers. Harv. Bus. Rev. 36, 37–66.
Forrester, J.W., 1961. Industrial Dynamics. The M.I.T. Press, Cambridge, MA.
Gao, D., Wang, N., He, Z., Jia, T., 2017. The bullwhip effect in an online retail supply chain: a perspective of price-sensitive demand based on the price discount in E-commerce. IEEE Trans. Eng. Manag. 64, 134–148. https://doi.org/10.1109/TEM.2017.2666265.
Geary, S., Disney, S.M., Towill, D.R., 2006. On bullwhip in supply chains—historical review, present practice and expected future impact. Int. J. Prod. Econ. 101, 2–18. https://doi.org/10.1016/j.ijpe.2005.05.009.
Glaser, B.G., Strauss, A.L., 1967. The Discovery of Grounded Theory: Strategies for Qualitative Theory. Aldine Transaction, New Brunswick, U.S.A.
Guan, W., Rehme, J., 2012. Vertical integration in supply chains: driving forces and consequences for a manufacturer's downstream integration. Supply Chain Manag.: Int. J. 17, 187–201. https://doi.org/10.1108/13598541211212915.
Haines, R., Hough, J., Haines, D., 2017. A metacognitive perspective on decision making in supply chains: revisiting the behavioral causes of the bullwhip effect. Int. J. Prod. Econ. 184, 7–20.
Hamilton, J.D., 2011. Historical Oil Shocks. National Bureau of Economic Research Working Paper Series. https://doi.org/10.3386/w16790. No. 16790.
Harrison, R., 1970. Regulation of well spacing in oil and gas production. Alberta Law Rev. 8, 357.
Hobbs, J.E., Young, L.M., 2000. Closer vertical co-ordination in agri-food supply chains: a conceptual framework and some preliminary evidence. Supply Chain Manag.: Int. J. 5, 131–143. https://doi.org/10.1108/13598540010338884.
Hussain, M., Drake, P.R., 2011. Analysis of the bullwhip effect with order batching in multi-echelon supply chains. Int. J. Phys. Distrib. Logist. Manag. 41, 797–814. https://doi.org/10.1108/09600031111166438.
Inkpen, A.C., Moffett, M.H., 2011. Global Oil and Gas Industry : Management, Strategy, and Finance. PennWell, Tulsa, OK, USA.
Isaksson, O.H.D., Seifert, R.W., 2016. Quantifying the bullwhip effect using two-echelon data: a cross-industry empirical investigation. Int. J. Prod. Econ. 171, 311–320. https://doi.org/10.1016/j.ijpe.2015.08.027.
Jacoby, D., 2010. The Oil Price 'bullwhip': Problem, Cost, Response. Oil & Gas Journal. PennWell Corporation, Tulsa, United States, Tulsa.

Jacoby, D., 2012. Optimal Supply Chain Management in Oil, Gas, and Power Generation. PennWell, Tulsa, OK, USA.

Kadivar, M., Akbarpour Shirazi, M., 2018. Analyzing the behavior of the bullwhip effect considering different distribution systems. Appl. Math. Model. 59, 319–340. https://doi.org/10.1016/j.apm.2018.01.028.

Kim, J.G., Chatfield, D., Harrison, T.P., Hayya, J.C., 2006. Quantifying the bullwhip effect in a supply chain with stochastic lead time. Eur. J. Oper. Res. 173, 617–636. https://doi.org/10.1016/j.ejor.2005.01.043.

Kurian, G.T., 2013. The AMA Dictionary of Business and Management. AMACOM, Saranac Lake, UNITED STATES.

Lee, H.L., Padmanabhan, V., Whang, S., 1997a. The bullwhip effect in supply chains. Sloan Manag. Rev. 38, 93–102. https://doi.org/10.1109/EMR.2015.7123235.

Lee, H.L., Padmanabhan, V., Whang, S., 1997b. Information distortion in a supply chain: the bullwhip effect. Manag. Sci. 43, 546–558. https://doi.org/10.1287/mnsc.43.4.546.

Li, Q., Disney, S.M., Gaalman, G., 2014. Avoiding the bullwhip effect using Damped Trend forecasting and the Order-Up-To replenishment policy. Int. J. Prod. Econ. 149, 3–16. https://doi.org/10.1016/j.ijpe.2013.11.010.

Ma, Y., Wang, N., Che, A., Huang, Y., Xu, J., 2013. The bullwhip effect under different information-sharing settings: a perspective on price-sensitive demand that incorporates price dynamics. Int. J. Prod. Res. 51, 3085–3116. https://doi.org/10.1080/00207543.2012.754551.

McCullen, P., Towill, D., 2002. Diagnosis and reduction of bullwhip in supply chains. Supply Chain Manag.: Int. J. 7, 164–179. https://doi.org/10.1108/13598540210436612.

McCutcheon, D.M., Meredith, J.R., 1993. Conducting case study research in operations management. J. Oper. Manag. 11, 239–256. https://doi.org/10.1016/0272-6963(93)90002-7.

Miles, M.B., Huberman, A.M., 1994. Qualitative Data Analysis: an Expanded Sourcebook. sage.

Miragliotta, G., 2006. Layers and mechanisms: a new taxonomy for the bullwhip effect. Int. J. Prod. Econ. 104, 365–381. https://doi.org/10.1016/j.ijpe.2005.10.001.

Moyaux, T., Chaib-draa, B., Amours, S.D., 2007. Information sharing as a coordination mechanism for reducing the bullwhip effect in a supply chain. IEEE Trans. Syst. Man Cybern. C Appl. Rev. 37, 396–409. https://doi.org/10.1109/TSMCC.2006.887014.

NEB, 2018. Western Canadian Crude Oil Supply, Markets, and Pipeline Capacity. National Energy Board, Alberta, Canada.

Osegowitsch, T., Madhok, A., 2003. Vertical integration is dead, or is it? Bus. Horiz. 46, 25–34. https://doi.org/10.1016/S0007-6813(03)00006-5.

Paik, S.K., Bagchi, P.K., 2007. Understanding the causes of the bullwhip effect in a supply chain. Int. J. Retail Distrib. Manag. 35, 308–324. https://doi.org/10.1108/09590550710736229.

Pastore, E., Alfieri, A., Zotteri, G., 2017. An empirical investigation on the antecedents of the bullwhip effect: evidence from the spare parts industry. Int. J. Prod. Econ. https://doi.org/10.1016/j.ijpe.2017.08.029.

Reinhard, E., Benjamin, S., Konrad, Z., 2017. Factors behind the 2014 Oil Price Decline. Bank of Canada Review-Autumn, pp. 1–13. *2017*.

Saccani, N., Johansson, P., Perona, M., 2007. Configuring the after-sales service supply chain: a multiple case study. Int. J. Prod. Econ. 110, 52–69. https://doi.org/10.1016/j.ijpe.2007.02.009.

Shan, J., Yang, S., Yang, S., Zhang, J., 2014. An empirical study of the bullwhip effect in China. Prod. Oper. Manag. 23, 537–551. https://doi.org/10.1111/poms.12034.

Simchi-Levi, D., Kaminsky, P., Simchi-Levi, E., Shankar, R., 2008. Designing and Managing the Supply Chain: Concepts, Strategies and Case Studies. Tata McGraw-Hill Education.

Sterman, J.D., 1989. Modeling managerial behavior: misperceptions of feedback in a dynamic decision making experiment. Manag. Sci. 35, 321–339. https://doi.org/10.1287/mnsc.35.3.321.

Sterman, J.D., 2006. Operational and behavioral causes of supply chain instability. In: Octavio, A., Carranza Torres, Felipe A., Morán, Villegas (Eds.), The Bullwhip Effect in Supply Chains: a Review of Methods, Components and Cases. Palgrave Macmillan UK.

Sucky, E., 2009. The bullwhip effect in supply chains—an overestimated problem? Int. J. Prod. Econ. 118, 311–322. https://doi.org/10.1016/j.ijpe.2013.11.010.

Taylor, D.H., 1999. Measurement and analysis of demand amplification across the supply chain. Int. J. Logist. Manag. 10, 55–70. https://doi.org/10.1108/09574099910805996;1910.1108/09574099910805996.

Taylor, D.H., 2000. Demand amplification: has it got us beat? Int. J. Phys. Distrib. Logist. Manag. 30, 515–533. https://doi.org/10.1108/09600030010372630.

Trapero, J.R., Fildes, R., Davydenko, A., 2011. Nonlinear identification of judgmental forecasts effects at SKU level. J. Forecast. 30, 490–508. https://doi.org/10.1002/for.1184.

Trapero, J.R., Pedregal, D.J., 2016. A novel time-varying bullwhip effect metric: an application to promotional sales. Int. J. Prod. Econ. 182, 465–471. https://doi.org/10.1016/j.ijpe.2016.10.004.

Voss, C., Tsikriktsis, N., Frohlich, M., 2002. Case research in operations management. Int. J. Oper. Prod. Manag. 22, 195–219. https://doi.org/10.1108/01443570210414329.

Wang, N., Lu, J., Feng, G., Ma, Y., Liang, H., 2016. The bullwhip effect on inventory under different information sharing settings based on price-sensitive demand. Int. J. Prod. Res. 54, 4043–4064. https://doi.org/10.1080/00207543.2016.1171418.

Wang, X., Disney, S.M., 2016. The bullwhip effect: progress, trends and directions. Eur. J. Oper. Res. 250, 691–701. https://doi.org/10.1016/j.ejor.2015.07.022.

Yin, R.K., 1994. Case Study Research: Design and Methods. Sage Publications, Thousand Oaks, CA.

Zhang, X., 2004. The impact of forecasting methods on the bullwhip effect. Int. J. Prod. Econ. 88, 15–27. https://doi.org/10.1016/S0925-5273(03)00128-2.

Zhang, X., Burke, G.J., 2011. Analysis of compound bullwhip effect causes. Eur. J. Oper. Res. 210, 514–526. https://doi.org/10.1016/j.ejor.2010.09.030.

Zuñiga, R., Wuest, T., Thoben, K.-D., 2015. Comparing mining and manufacturing supply chain processes: challenges and requirements. Prod. Plan. Control 26, 81–96. https://doi.org/10.1080/09537287.2013.855335.